Anomalous Is Not Malicious: A Provable Performance Ceiling for Unsupervised NIDS and a Two-Stage Architecture for Zero-Day Intrusion Detection

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***Abstract*—Unsupervised network intrusion detection systems (NIDS) consistently plateau at 55–70% recall on real-world benchmarks, a degradation widely attributed to parameter misconfiguration. We prove this ceiling is *structural*: for any threshold-based detector, the maximum achievable Youden index ($\mathrm{TPR} - \mathrm{FPR}$) equals the Kolmogorov–Smirnov (KS) distance between the class-conditional anomaly-score distributions (Theorem 1). On the CTU-13 botnet dataset, Isolation Forest attains $\mathrm{KS} = 0.445$; no threshold calibration can exceed this bound. The root cause is the *anomaly–malice mismatch*: unsupervised detectors measure statistical rarity, whereas stealthy attacks are engineered to appear statistically normal. Replacing the anomaly notion — isolation by reconstruction, via an autoencoder trained exclusively on normal traffic — raises the ceiling to $\mathrm{KS} = 0.860$ (F_1: $0.617 \to 0.928$), and the deployed detector operates *at* its theoretical ceiling (balanced accuracy 0.929 vs. bound 0.930). Cross-dataset evaluation on UNSW-NB15, however, shows no single anomaly notion generalises across attack families. We therefore derive a closed-form law for OR-fusion of conditionally independent detectors (Theorem 2): miss rates multiply. The law predicts the fused recall of our dual-head detector (flow autoencoder $\cup$ temporal autoencoder) to three decimal places (predicted 0.488, measured 0.491; miss-redundancy $\rho_{\mathrm{miss}} = 0.994$), lifting Stage-1 recall from 8.6% to 49.1% and, on hidden attack families, Backdoor from 8% to 41% and Analysis from 3% to 35%. Finally, we replace the arbitrary percentile threshold with an extreme-value-theoretic threshold (Theorem 3): a generalized Pareto tail fit ($\hat{\xi} = 1.12$, infinite variance) yields calibrated false-positive rates where percentile thresholds overshoot by $3.8\times$. The resulting two-stage architecture — fused zero-day net, gradient-boosted known-attack classifier, and a zero-day candidate queue — achieves $F_1 = 0.945$ on CTU-13 with 87% false-positive reduction, and in a controlled simulation withholding four attack families from the classifier, routes all four to the zero-day queue at 71% queue precision with zero misclassifications of hidden attacks as known types.**

***Index Terms*—Network intrusion detection, anomaly detection, zero-day attacks, Kolmogorov–Smirnov distance, extreme value theory, detector fusion, Isolation Forest, autoencoder, CTU-13, UNSW-NB15.**

I. Introduction

SIGNATURE-BASED intrusion detection fails categorically against zero-day attacks: no signature exists for an attack no one has seen. Anomaly-based NIDS promise the alternative — model normal traffic, flag deviations — yet two decades of published results place unsupervised recall stubbornly in the 55–70% band [3], [5]. The standard prescriptions (tune the contamination parameter, add features, deepen the model) do not move this number. This paper explains why, proves the limitation formally, and presents a validated architecture that circumvents it.

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Our contributions:

1) **A provable ceiling (Sec. III).** For threshold detectors, $\sup_t J(t) = \mathrm{KS}$ exactly. Corollaries give a balanced-accuracy ceiling $(1 + \mathrm{KS})/2$ and a recall bound at any false-alarm budget. On CTU-13 our autoencoder *attains* its ceiling, proving threshold optimisation is exhausted and only representation change can help.
2) **A fusion law (Sec. III).** Under conditional independence, OR-fused detector misses multiply. The law is verified on real traffic to three decimal places and supplies a closed-form criterion for when fusing two detectors is beneficial.
3) **Parameter-free thresholds (Sec. III).** A peaks-over-threshold GPD fit converts any target false-positive budget into a calibrated threshold, including budgets below the empirical resolution of the training data.
4) **A validated two-stage system (Sec. V).** Dual-head Stage 1 (flow + temporal autoencoders, fused per the law), a supervised Stage 2 over the flagged pool, and an explicit zero-day candidate queue, evaluated with a hold-out zero-day simulation on UNSW-NB15.

II. Related Work

Isolation Forest [1] and extensions [8]; LOF [2]; autoencoder NIDS [6], [7]; EVT for anomaly thresholds [9]; CTU-13 [3] and UNSW-NB15 [4] benchmarks. Our work differs by diagnosing the ceiling formally before proposing remedies, and by validating each closed-form prediction on real traffic.

III. Theoretical Framework

Let S be a scalar anomaly score (higher = more anomalous), $Y \in \{0,1\}$ the true class (1 = attack), and $F_A(t) = P(S \le t \mid Y{=}1)$, $F_N(t) = P(S \le t \mid Y{=}0)$ the class-conditional CDFs. A threshold detector decides "attack" iff $S > t$; its operating point is $\mathrm{TPR}(t) = 1 - F_A(t)$, $\mathrm{FPR}(t) = 1 - F_N(t)$, and its Youden index is $J(t) = \mathrm{TPR}(t) - \mathrm{FPR}(t)$.

A. The KS Ceiling

Theorem 1 (KS ceiling)**.** *For any threshold detector on S, in either orientation,*

$$\sup_t |J(t)| \;=\; \sup_t |F_N(t) - F_A(t)| \;=\; \mathrm{KS}(F_A, F_N),$$

and the bound is invariant under any monotone transformation of S.

Proof. $J(t) = (1 - F_A(t)) - (1 - F_N(t)) = F_N(t) - F_A(t)$; take suprema over t and over the two orientations. Monotone maps preserve both CDFs' level sets, hence KS. □

Corollary 1 (Balanced-accuracy ceiling)**.** $\max_t \mathrm{BA}(t) = \frac{1}{2}\big(1 + \mathrm{KS}\big)$.

Corollary 2 (Recall at a false-alarm budget)**.** $\mathrm{TPR}(t) \leq \mathrm{FPR}(t) + \mathrm{KS}$ *for every t; hence at budget $\mathrm{FPR} \leq \alpha$, recall is at most $\alpha + \mathrm{KS}$.*

Corollary 3 (AUC bound)**.** $\mathrm{AUC} \leq \frac{1}{2} + \mathrm{KS}$.

Scope. For arbitrary measurable decision regions the bound is the total variation distance $\mathrm{TV} \geq \mathrm{KS}$, with equality under monotone likelihood ratio; deployed detectors are threshold rules, so KS is the operative ceiling.

Empirical validation (CTU-13, held-out test). The empirical $\sup_t J(t)$ equals the two-sample KS statistic to machine precision for both detectors (difference $\leq 5.6 \times 10^{-17}$). Corollary 1: autoencoder $\mathrm{KS} = 0.863$ gives ceiling 0.931; observed balanced accuracy 0.929 — *the deployed detector operates at its theoretical ceiling*. Corollary 2: Isolation Forest ($\mathrm{KS} = 0.457$) cannot exceed 50.7% recall at a 5% false-alarm budget, explaining the widely reported plateau.

B. OR-Fusion of Conditionally Independent Detectors

Let $B_i = \{S_i > t_i\}$, $i = 1, 2$, be the flag events of two detectors with operating points $(\mathrm{TPR}_i, \mathrm{FPR}_i)$, and assume conditional independence (CI) given each class: $P(B_1 \cap B_2 \mid Y{=}y) = P(B_1 \mid y)P(B_2 \mid y)$.

Theorem 2 (Fusion law)**.** *Under (CI), the OR-fused detector satisfies:*

(a) Miss-product law: $1 - \mathrm{TPR}_{\mathrm{OR}} = (1 - \mathrm{TPR}_1)(1 - \mathrm{TPR}_2)$*;*
(b) Subadditive false alarms: $\mathrm{FPR}_{\mathrm{OR}} = 1 - (1 - \mathrm{FPR}_1)(1 - \mathrm{FPR}_2) \leq \mathrm{FPR}_1 + \mathrm{FPR}_2$*;*
(c) Fusion-benefit criterion: $J_{\mathrm{OR}} > J_1 \iff (1 - \mathrm{TPR}_1)\,\mathrm{TPR}_2 > (1 - \mathrm{FPR}_1)\,\mathrm{FPR}_2$.

Proof. (a),(b): apply (CI) to the complements — a union miss is a simultaneous miss. (c): $J_{\mathrm{OR}} - J_1 = (1 - \mathrm{TPR}_1)\mathrm{TPR}_2 - (1 - \mathrm{FPR}_1)\mathrm{FPR}_2$ by direct expansion. □

Proposition 1 (Mixture limit)**.** *If the attack population is a mixture in which subfamily k is invisible to the other head (likelihood ratio $\equiv 1$), the Neyman–Pearson optimal region degenerates to the union $B_1 \cup B_2$: OR-fusion is exactly optimal for heterogeneous attack families.*

We diagnose (CI) with the miss-redundancy coefficient $\rho_{\mathrm{miss}} = P(\bar{B}_1 \cap \bar{B}_2 \mid Y{=}1) \,/\, \big[P(\bar{B}_1 \mid Y{=}1)P(\bar{B}_2 \mid Y{=}1)\big]$ ($\rho = 1$: independent blind spots) and the conditional mutual information $I(S_1; S_2 \mid Y)$ estimated by within-class quantile binning.

Empirical validation (UNSW-NB15, flow-AE head vs. temporal-AE head). Head operating points $\mathrm{TPR}_1 = 0.086$, $\mathrm{TPR}_2 = 0.440$ (both at 95th-percentile thresholds). The law predicts union recall $1 - (1 - 0.086)(1 - 0.440) = 0.488$; measured: 0.491. Overall $\rho_{\mathrm{miss}} = 0.994$ and $I(S_1; S_2 \mid Y) = 0.28$ bits — the two anomaly notions are empirically conditionally independent. Per-family $\rho_{\mathrm{miss}} \in [1.00, 1.13]$. The fusion criterion holds with a $6\times$ margin. Fused Stage-1 recall on hidden families: Backdoor $8\% \to 41\%$, Analysis $3\% \to 35\%$.

C. Calibrated Thresholds via Extreme Value Theory

Theorem 3 (Pickands–Balkema–de Haan, applied)**.** *If the normal-score distribution lies in the maximum domain of attraction of an extreme value law, the excess distribution over a high threshold u converges uniformly to a generalized Pareto distribution $G_{\xi,\sigma}(x) = 1 - (1 + \xi x/\sigma)^{-1/\xi}$. Consequently, with $p_u = P(S_N > u)$ estimated empirically and $(\hat{\xi}, \hat{\sigma})$ fitted by maximum likelihood on the excesses, the threshold achieving a target false-positive rate q is*

$$t_q \;=\; u + \frac{\hat{\sigma}}{\hat{\xi}}\Big[\big(p_u/q\big)^{\hat{\xi}} - 1\Big] \qquad (\hat{\xi} \neq 0).$$

Empirical validation (UNSW-NB15 flow-AE, $n = 56{,}000$ training normals, $u =$ 90th percentile). The fitted shape is $\hat{\xi} = 1.12$: the reconstruction-error tail is so heavy its variance is infinite, which is precisely why percentile thresholds miscalibrate. At target $q = 10^{-3}$ the GPD threshold realises $\mathrm{FPR} = 4.9 \times 10^{-4}$ on held-out normals while the percentile threshold realises 3.8×10^{-3} ($3.8\times$ over budget). The GPD extrapolates analytically to budgets below the $1/n$ resolution of the data, removing the last free parameter from Stage 1.

IV. Datasets and Experimental Protocol

CTU-13 [3]: 38,898 botnet flows, 53,314 normal, 57 CICFlowMeter features; balanced 6k/6k samples, 70/30 stratified split. UNSW-NB15 [4]: 175,341 train / 82,332 test flows, nine attack families, 39 Argus features. All preprocessing (median imputation, $\log(1{+}x)$ on skewed columns, standardisation) fitted on training partitions only.

V. The Two-Stage Architecture

Stage 1: dual autoencoder heads (flow features; temporal/behavioural features), each trained on normal traffic only, thresholds set by Theorem 3, fused by OR per Theorem 2. Stage 2: gradient-boosted classifier over the flagged pool; flows below confidence $\tau = 0.6$ enter the *zero-day candidate queue*.

Current single-head results: CTU-13 pipeline $F_1 = 0.945$ with an 87% false-positive reduction over Stage 1 alone; zero-day simulation withholding Backdoor, Shellcode, Analysis and Worms from Stage 2 routes all four families to the queue (71% queue precision, zero hidden-family misclassifications). The fused-head end-to-end evaluation is the final experiment before submission.

VI. Discussion and Limitations

VII. Conclusion

References

[1] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proc. IEEE ICDM*, 2008, pp. 413–422.

[2] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, "LOF: Identifying density-based local outliers," in *Proc. ACM SIGMOD*, 2000, pp. 93–104.

[3] S. Garcia, M. Grill, J. Stiborek, and A. Zunino, "An empirical comparison of botnet detection methods," *Computers & Security*, vol. 45, pp. 100–123, 2014.

[4] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems," in *Proc. MilCIS*, 2015, pp. 1–6.

[5] N. Moustafa and J. Slay, "The evaluation of network anomaly detection systems," *Information Security Journal*, vol. 25, no. 1–3, pp. 18–31, 2016.

[6] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An ensemble of autoencoders for online network intrusion detection," in *Proc. NDSS*, 2018.

[7] S. Zavrak and M. Iskefiyeli, "Anomaly-based intrusion detection from network flow features using variational autoencoder," *IEEE Access*, vol. 8, pp. 108346–108358, 2020.

[8] S. Hariri, M. C. Kind, and R. J. Brunner, "Extended isolation forest," *IEEE Trans. Knowl. Data Eng.*, vol. 33, no. 4, pp. 1479–1489, 2021.

[9] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouët, "Anomaly detection in streams with extreme value theory," in *Proc. ACM KDD*, 2017, pp. 1067–1075.